A Unified Box-Free Network for Joint Instance Segmentation and Pose Estimation of Laboratory Mice

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Abstract. Automated tracking of morphology and laboratory-mice posture usually runs instance segmentation and pose estimation as two separate pipelines, which wastes computation, or uses proposal-driven detectors that blur object boundaries through coarse, fixed-resolution mask prediction. To address these limitations, we propose **HybridMTLNet**, a unified, box-free multi-task learning framework for simultaneous multi-instance segmentation and keypoint localization in a single forward pass. The architecture uses a shared backbone encoder, a dense per-pixel segmentation decoder, and a deconvolutional pose head. During training, we apply Hungarian matching on mask IoU to assign predictions to ground-truth instances in an order-invariant manner. We evaluate HybridMTLNet on 3,710 manually annotated images spanning eight mouse behavioral tests. Averaged over three random seeds, our model achieves a Percentage of Correct Keypoints (PCK@0.05) of 0.9605 and a mean Intersection over Union (mIoU) of 0.8083, matching or surpassing dedicated single-task baselines. A controlled ablation shows that the multi-task model nearly matches its single-task counterparts of the same architecture—preserving segmentation accuracy at only a small cost in pose—while roughly halving the computation.

Keywords: Multi-Task Learning · Computer Vision · Image Understanding · Instance Segmentation · Pose Estimation · Hungarian Matching · Laboratory Mice · Mice Behavior Analysis

1 Introduction

Automated tracking of morphology and laboratory-mice posture plays an important role in modern behavioral neuroscience and biomedical research, where accurate monitoring lets researchers quantify complex behaviors, evaluate phenotypic variation, and assess pharmacological interventions. Two computer-vision tasks underpin this analysis: instance segmentation, which outlines each animal's body, and pose estimation, which locates its joints. In practice, however, these usually run as two separate pipelines, which wastes computation, or rely

on proposal-driven detectors that blur object boundaries through coarse, fixed-resolution mask prediction.

To address these limitations, we propose HybridMTLNet, a unified, box-free multi-task learning framework for simultaneous multi-instance segmentation and keypoint localization in a single forward pass. The architecture uses a shared backbone encoder, a dense per-pixel segmentation decoder, and a deconvolutional pose head. During training, we apply Hungarian matching on mask IoU to assign predictions to ground-truth instances in an order-invariant manner, removing the need for bounding-box proposals. Our goal is not a multi-task accuracy gain—which our ablation does not find—but a single network that preserves segmentation accuracy at only a small cost in pose while roughly halving the computation.

The main contributions of this work are summarized as follows:

- **Unified box-free architecture.** A single network that performs multi-instance segmentation and keypoint localization in one forward pass, removing the redundant computation of separate networks and the bounding boxes of proposal-driven detectors.
- **Order-invariant training.** Hungarian matching on mask IoU that assigns predictions to ground-truth instances, resolving the label-switching ambiguity of joint training.
- **Controlled empirical evaluation.** On 3,710 manually annotated images spanning eight mouse behavioral tests, HybridMTLNet reaches a PCK@0.05 of 0.9605 and an mIoU of 0.8083, matching or surpassing dedicated single-task baselines while roughly halving the computation.

2 Related Work

2.1 Estimating Animal Visual Pose

Locating anatomical landmarks in images has progressed from classical part-based models to deep learning. Early methods, such as the deformable part models of Yang and Ramanan [23], used mixtures-of-parts to model body configurations and popularized the Percentage of Correct Keypoints (PCK) as a standard metric. Newell et al. [13] reshaped the field with the Stacked Hourglass Network, which casts keypoint localization as Gaussian heatmap regression. HRNet [18] later refined this paradigm, maintaining high-resolution feature representations throughout the network. For biological research, open-source markerless frameworks such as DeepLabCut [11] and SLEAP [15] adapted these heatmap-regression methods to laboratory-animal tracking, with SLEAP extending to multi-animal social settings through Part Affinity Fields (PAFs). However, these frameworks handle only sparse keypoints and cannot capture fine object shape or boundaries.

2.2 Dense Instance Segmentation

Top-down, proposal-driven architectures have long been the dominant paradigm for instance segmentation. Mask R-CNN [3] pioneered the detect-then-segment approach and still sets the standard. Recent iterations, such as YOLO-Seg [6], optimize these pipelines for real-time one-stage deployment. Despite their efficiency, these frameworks crop each detected instance to a region of interest and predict its mask on a fixed, low-resolution grid (for example, 28×28 in Mask R-CNN). Upsampling this coarse mask back to full resolution blurs fine object boundaries. To relieve these artifacts, proposal-free and box-free frameworks have emerged. In particular, SOLOv2 [20] formulates instance segmentation as a location-category prediction problem that directly yields high-fidelity dense masks from continuous feature grids without bounding-box interventions. However, these box-free models target only mask generation and do not support extra tasks such as keypoint regression or joint pose estimation.

2.3 Multi-Task Learning of Representations

Multi-task learning (MTL) trains a single shared backbone across multiple objective functions to learn generalized representations [2]. Joint segmentation and pose estimation have been integrated into unified networks, such as Parsing R-CNN [22] and UniPose [1], for human vision analysis. However, joint optimization is often hampered by gradient conflicts, also known as task interference or negative transfer, where competing objectives prevent the shared weights from reaching the best solution for each task [7,17]. Gradient surgery techniques (e.g., PCGrad [24]) address this at the optimization stage. In automated animal behavioral analysis, however, multi-task tracking has traditionally relied on modular pipelines.

Frameworks such as SIPEC [10], STCS [19], and InteBOMB [25] chain together independent models—for example, Mask R-CNN for instance localization followed by separate networks for identification and pose estimation. Because these models share no internal features, they repeat feature extraction and incur high latency. Recently, unified single-network alternatives such as YOLO-SP [5] and MT-SRNet [9] have been proposed, performing multi-task tracking in a single framework. However, they remain detection-based and depend on either bounding-box proposals or rotated rectangles. Moreover, these models are mostly developed for agricultural settings such as crop harvesting and livestock (e.g., pigs) in open environments, and are not well adapted to the spatial constraints and structural ambiguities of small laboratory-rodent tracking.

To overcome these limitations, HybridMTLNet is a proposal-free, box-free, multi-task network that tracks laboratory mice in a single forward pass. Our framework unifies instance segmentation and keypoint localization via a shared encoder and dynamic Hungarian matching, eliminating the redundant computational cost of modular pipelines and the boundary artifacts of proposal-based mask heads.

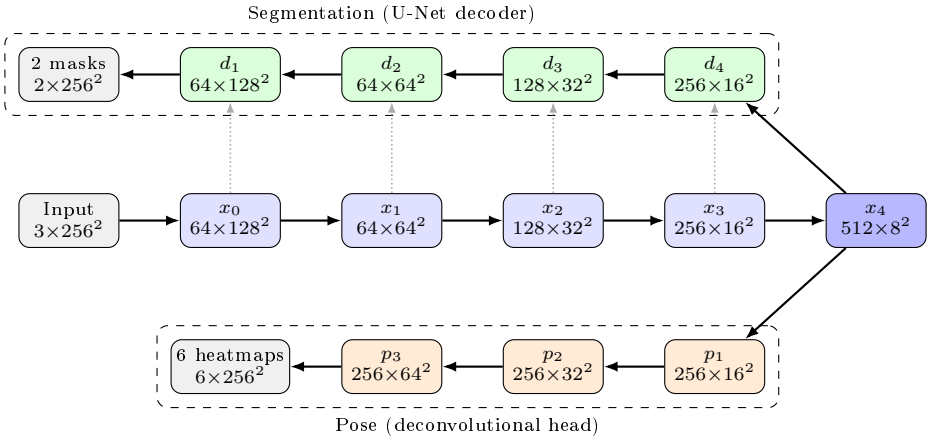


Fig. 1. HybridMTLNet architecture (channels × spatial dimensions are indicated per block). A shared ResNet-34 encoder extracts multi-scale features x_0–x_4. The bottleneck representation x_4 feeds two parallel branches: a U-Net-based segmentation decoder (d_4–d_1, with dotted lines denoting skip connections from the encoder) yielding two dense instance masks, and a deconvolutional pose head (p_1–p_3) yielding six keypoint heatmaps (3 keypoints × 2 instances). Both outputs are restored to the original 256×256 resolution.

3 Methodology

3.1 Dataset

The dataset comprises 3,710 images collected across eight distinct laboratory behavioral tests: the Morris water maze, dark/light box, novel object recognition, forced swim, Y-maze, mirror box, elevated plus maze, and tail suspension. The images contain either one or two mice (64.4% single-mouse and 35.6% two-mouse configurations) and are manually annotated with per-instance binary segmentation masks and three skeletal keypoints per animal (the nose, shoulder, and tail base). All annotations were completed using the Label Studio tool, where each label underwent cross-verification by a second annotator, resolving any inconsistencies via consensus. Since the framework requires no bounding boxes, annotations consist of masks and keypoints only. For experimental evaluation, the dataset is partitioned into training (2,968), validation (371), and testing (371) sets according to an 80/10/10 ratio. While the source photographs feature mixed resolutions (640×360 and 1280×720), all subsequent models are trained and evaluated at a standardized spatial resolution of 256×256.

3.2 Architecture

As illustrated in Fig. 1, HybridMTLNet performs instance segmentation and keypoint estimation in a single forward pass. The network consists of three components: a shared encoder, a segmentation decoder, and a pose head.

Shared Encoder. The backbone is a ResNet-34 [4] pretrained on ImageNet. Sharing this encoder across both tasks eliminates redundant feature extraction. Features are extracted at four distinct resolutions and passed to the decoder via skip connections, preserving fine spatial detail.

Segmentation Decoder. The segmentation branch is a dense per-pixel decoder [16] that progressively upsamples the low-resolution bottleneck features via bilinear interpolation. Skip-connected encoder features restore spatial detail at each resolution level. The output is a 2-channel mask ($2 \times H \times W$), where each channel corresponds to one mouse. By predicting at full 256×256 resolution—rather than upsampling from a low-resolution RoI grid—the decoder produces precise mask boundaries.

Pose Head. For keypoint estimation, we use heatmap regression rather than directly regressing (x, y) coordinates via fully connected layers, an approach that discards spatial structure. The pose head consists of three transposed-convolution layers that progressively upsample the shared features, followed by a 1×1 convolution to produce a $6 \times H \times W$ heatmap (3 keypoints $\times$ 2 instances). Each channel encodes the likelihood that the keypoint lies at each spatial location; at inference, the coordinates are read as the argmax of the corresponding channel.

The two output heads are fixed-width—a two-channel mask and six heatmaps—so the network handles at most two instances per image, matching our dataset. For single-mouse images, the surplus mask and heatmap channels are supervised against empty (all-zero) targets. The network does not always leave this slot empty—it tends to replicate the single instance rather than invent a new one—but, as the replicas overlap, a simple suppression step removes them at inference without affecting the matched-instance metrics (Section 5.1).

3.3 Loss Function

We train HybridMTLNet with a single multi-task objective. The overall loss ($\mathcal{L}_{\text{total}}$) is the weighted sum of the segmentation loss ($\mathcal{L}_{\text{seg}}$) and the pose loss ($\mathcal{L}_{\text{pose}}$):

$$\mathcal{L}_{\text{total}} = \lambda_{\text{seg}}\,\mathcal{L}_{\text{seg}} + \lambda_{\text{pose}}\,\mathcal{L}_{\text{pose}} \tag{1}$$

where λ_{seg} and λ_{pose} are scalar weights controlling the relative gradient magnitudes of the two tasks.

Segmentation Loss. The segmentation loss combines Binary Cross-Entropy (BCE) and Dice loss [12] for supervision at both the pixel and structure levels:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{BCE}} + \left(1 - \frac{2\sum p\,g + \epsilon}{\sum p + \sum g + \epsilon}\right) \tag{2}$$

where $\epsilon = 1$ is a smoothing constant. BCE penalizes pixel-wise misclassifications between the predicted and ground-truth masks. The Dice term measures the overall foreground overlap, preventing the network from collapsing to an all-background prediction—a failure mode common in cage scenes where the background dominates.

Pose Loss. The pose loss is the Mean Squared Error (MSE) between the predicted heatmap ($\hat{H}$) and the target Gaussian heatmap (H), averaged over all $K = 6$ keypoint channels and the $H \times W$ pixels:

$$\mathcal{L}_{\text{pose}} = \frac{1}{KHW} \sum_{k=1}^{K} \|\hat{H}_k - H_k\|_2^2 \tag{3}$$

The squared penalty encourages precise keypoint localization.
Loss Weight Balancing. The weights are set to $\lambda_{\text{seg}} = 1$ and $\lambda_{\text{pose}} = 500$. The target heatmaps are highly sparse—mostly zeros except at small Gaussian peaks—so the magnitude of $\mathcal{L}_{\text{pose}}$ is intrinsically two to three orders of magnitude smaller than $\mathcal{L}_{\text{seg}}$. Without this scaling, segmentation dominates the gradient and the pose head trains poorly. Section 5.3 analyzes this choice empirically.

3.4 Order-Invariant Instance Matching

HybridMTLNet is a box-free model (a 2-channel mask and a 6-channel heatmap), and its output channels are unordered. In one image, the network may place the left mouse in the first channel, whereas for another image it may place the right mouse in the first channel. This phenomenon is termed permutation symmetry, or label switching.

We overcome this by performing a three-step alignment at each training step. **Step 1 — Construct a cost matrix.** Let N_{pred} and N_{gt} be the numbers of predicted and ground-truth instances, respectively. We construct a cost matrix C of size $N_{\text{pred}} \times N_{\text{gt}}$, where each entry $C_{i,j}$ represents the dissimilarity between prediction i and ground-truth instance j, defined as one minus their mask overlap (IoU):

$$C_{i,j} = 1 - \text{IoU}(\hat{S}_i, S_j) \tag{4}$$

Here $\hat{S}_i$ is the predicted mask binarized at a 0.5 sigmoid threshold and S_j the ground-truth instance mask. The larger the overlap, the higher the IoU and the lower the cost. **Step 2 — Find the best matching.** We use the Hungarian algorithm (linear sum assignment) [8] to find the one-to-one matching with the lowest total cost, which is equivalent to the largest total mask overlap. **Step 3 — Reorder the ground truth.** Using this matching, we reorder the ground-truth masks and heatmaps to match the output channels of the model, and only then compute $\mathcal{L}_{\text{total}}$. This guarantees that the gradient is consistent across training steps rather than contradictory. The assignment itself is non-differentiable and detached from the computation graph, so gradients flow through the loss but not through the assignment.

3.5 Data Augmentation

To enhance generalization and robustness, we apply online data augmentation during training. The following transformations are applied randomly to each training sample with probability $p = 0.5$: horizontal flipping, vertical flipping,

random rotation within $\pm 30°$, and color jittering (brightness in $[0.7, 1.3]$ and contrast in $[0.8, 1.2]$). To maintain cross-task consistency between the two output heads, all geometric transformations (flipping and rotation) are applied identically to the input image, the ground-truth instance masks, and the skeletal keypoint coordinates. By contrast, color jittering is applied only to the input image, ensuring that photometric changes do not distort any spatial annotation.

4 Experimental Setup

4.1 Implementation Details

All models were implemented in PyTorch and trained on a single NVIDIA GeForce RTX 5060 GPU with 8 GB of memory. For HybridMTLNet, we used the Adam optimizer with an initial learning rate of 1×10^{-4}. Rather than a fixed schedule, we let the rate adapt: the ReduceLROnPlateau scheduler halves the rate whenever the validation loss stops improving for 5 consecutive epochs. Training ran for 100 epochs at a batch size of 16, with every input image resized to 256×256.

For reproducibility, we forced cuDNN into deterministic mode and reused the same data split across all runs. Because GPU training is not perfectly deterministic, we report results as the mean $\pm$ standard deviation over three random seeds (42, 43, and 44) for HybridMTLNet, the YOLO baselines, and the ablation configurations. The loss-weight sweep (Table 2) and the training curves (Fig. 4) show a single representative run at the chosen 1:500 ratio.

The baselines follow their own standard recipes. The YOLO-based models [6] were fine-tuned for 100 epochs with their default pipelines. Mask R-CNN [3], by contrast, was trained as a single representative run with SGD for the same duration under the same data augmentation as ours, owing to its substantially higher training cost. For multi-animal DeepLabCut [11], we ran 10,000 iterations in static-image mode and omitted the downstream identity-assembly stage, since our evaluation uses isolated, non-consecutive frames rather than video.

4.2 Evaluation Metrics

To compare multi-task and single-task designs fairly, we use two standard metrics.

Mean Intersection over Union (mIoU). For instance segmentation, it measures the per-instance IoU across the dataset:

$$\text{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \frac{|P_i \cap G_i|}{|P_i \cup G_i|} \tag{5}$$

For fairness, we use Hungarian matching on the spatial overlap to match predicted masks to ground-truth masks. Any unmatched ground-truth mask is scored as IoU 0.

Table 1. Test set performance. YOLO baselines and HybridMTLNet are reported as the mean ± standard deviation over three random seeds; Mask R-CNN is reported from a single representative run due to its high training cost, using the same data split, input resolution, and evaluation protocol. DeepLabCut is included as a reference baseline. Best results in **bold**; "—" indicates the model does not perform that task.

Model	Tasks	mIoU	PCK@0.05
YOLO26m-Seg	Seg only	0.5497 ± 0.0181	—
Mask R-CNN	Seg only	0.7854	—
YOLO26m-Pose	Pose only	—	0.8823 ± 0.0053
DeepLabCut (maDLC)	Pose only	—	0.4098
HybridMTLNet (Ours)	**Seg + Pose**	$\mathbf{0.8083 \pm 0.0012}$	$\mathbf{0.9605 \pm 0.0027}$
Δ vs. best evaluated single-task		+0.0229	+0.0782

Percentage of Correct Keypoints (PCK@0.05). For pose estimation, it follows the standard PCK protocol popularized by Yang and Ramanan [23] and adopted by Mathis et al. (DeepLabCut) [11], which considers a predicted keypoint correct if its Euclidean distance to the ground-truth annotation is smaller than a specified threshold. The threshold in this study was set to 5% of the image dimension, equivalent to a radius of 12.8 pixels at a resolution of 256×256. Keypoints tagged as invisible or occluded (annotated at source) are excluded from the assessment set.

4.3 Baseline Models

We compare HybridMTLNet with four well-known single-task networks.
Segmentation baselines: YOLO26m-Seg [6] and Mask R-CNN [3] (ResNet-50-FPN backbone, pretrained on COCO). These models are trained only for instance-level segmentation.
Pose estimation baselines: YOLO26m-Pose [6] and multi-animal DeepLabCut (maDLC) [11]. These models only locate joint coordinates. We exclude SLEAP [15], which targets multi-animal pose tracking in video and outputs no dense instance masks, so it is not directly comparable on our joint task.

All of these baselines are single-task. Obtaining both outputs therefore requires running two separate models, which is far more costly than HybridMTLNet's single forward pass.

5 Results and Discussion

5.1 Quantitative Comparison

We benchmark HybridMTLNet against four established single-task baselines. Table 1 reports the results for both the instance segmentation and keypoint estimation tasks.

In the instance segmentation task, HybridMTLNet attains an mIoU of 0.8083, above the strongest segmentation baseline (the single-run Mask R-CNN, 0.7854). Because our dense per-pixel decoder predicts at the full 256×256 resolution, it avoids the proposal cropping and fixed-resolution mask prediction that limit boundary precision in proposal-based mask heads. Our ablation (Section 5.4) confirms that this segmentation quality stems from the architecture, not from multi-task learning synergy.

For pose estimation, HybridMTLNet achieves a Percentage of Correct Keypoints (PCK@0.05) of 0.9605, an absolute gain of 0.0782 over YOLO26m-Pose. We include multi-animal DeepLabCut (maDLC) as a domain-standard reference, which scores 0.4098. This figure reflects the difficulty of our evaluation protocol rather than a weakness of maDLC: because our test set consists of isolated, non-consecutive frames, maDLC cannot exploit the temporal tracking and identity-assembly stages it is designed around. We therefore treat YOLO26m-Pose as the primary pose baseline and do not claim superiority over maDLC in its intended video setting.

Because HybridMTLNet emits a fixed two-channel mask, we also report instance-level precision and recall, treating each non-empty channel as a detection matched to ground truth at $\mathrm{IoU} > 0.5$. The raw output reaches 0.95 recall but only 0.64 precision: on single-mouse images the unused channel usually duplicates the instance. A simple mask-overlap suppression ($\mathrm{IoU} > 0.5$) restores precision to 0.93 with recall and matched-instance mIoU essentially unchanged. These duplicates overlap the true instance and are trivially removed; they do not affect the matched-instance metrics in Table 1.

In summary, each baseline is single-task and needs a separate pipeline per metric, whereas HybridMTLNet matches or exceeds them on both metrics in a single forward pass.

5.2 Qualitative Analysis

Figure 2 shows HybridMTLNet's predictions on test images from each of the eight behavioral tests. In all cases, our approach produces per-instance segmentation masks and three-keypoint skeletons simultaneously for up to two subjects in a single forward pass, indicating that the shared encoder learns features that generalize across diverse scenarios. Figure 3 complements this with the hardest cases (lowest mIoU/PCK): errors concentrate in low-contrast arenas, reflective scenes, and self-occluded postures rather than in a single architectural weakness, motivating contrast-aware augmentation as future work.

5.3 Loss-Weight Selection

We select the pose-loss weight λ_{pose} on the validation set, training the joint model across a range of seg:pose ratios (Table 2). Validation mIoU is stable across all ratios (0.785–0.796), confirming segmentation is insensitive to the pose weight. Validation PCK, in contrast, rises sharply with λ_{pose}—from 0.741 at 1:1 to 0.958 at 1:400—then saturates. We adopt 1:500 as a balanced operating point: its PCK

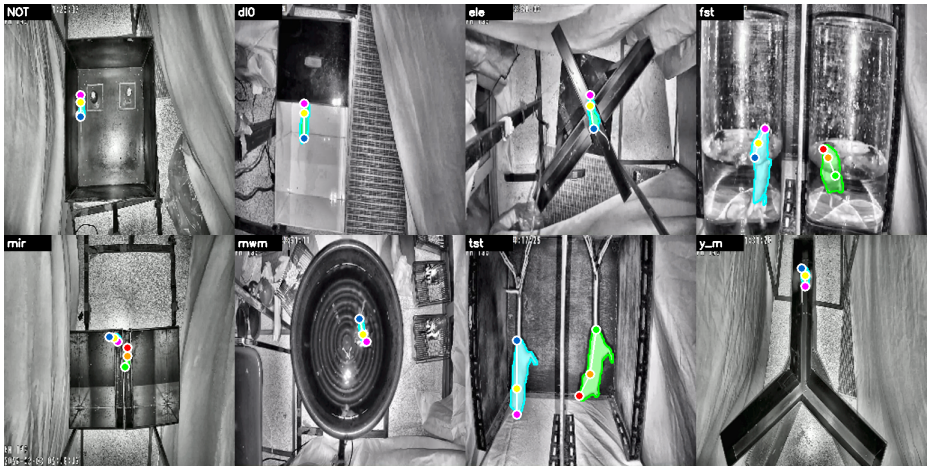


Fig. 2. Qualitative results across the eight behavioral tests. Each tile shows HybridMTLNet's per-instance segmentation masks and keypoint skeletons (nose–shoulder–tail) on a held-out test image.

Table 2. Loss-weight ratio (seg:pose) on the joint model; validation-set scores for a single representative run, used to select λ_{pose}. mIoU stays stable while PCK rises with the pose weight then saturates; we adopt 1:500, whose 3-seed test mean is reported in Table 1.

Ratio (seg:pose)	mIoU	PCK@0.05
1:1	0.7867	0.7413
1:100	0.7854	0.8031
1:400	0.7955	0.9580
1:500	0.7948	0.9595
1:1000	0.7915	0.9611

(0.9595) sits in the saturated region while its mIoU (0.7948) stays high. Pushing the weight to 1:1000 yields a negligible PCK gain (0.9611) but no benefit to segmentation; we prefer 1:500 to avoid sacrificing either task. The test set is reserved for final evaluation (Table 1).

5.4 Single-Task Ablation Study

To separate the effect of multi-task learning from the architecture itself, we run a controlled ablation. Using the same network, we compare three settings obtained by tuning the loss weights ($\lambda_{\text{seg}}, \lambda_{\text{pose}}$): (1) a segmentation-only network, (2) a pose-only network, and (3) our joint multi-task setup (Joint MTL). A quantitative breakdown is shown in Table 3.



Fig. 3. Failure cases: the worst test images by per-instance mIoU and PCK. Predicted masks and keypoint skeletons are overlaid; white crosses mark ground-truth keypoints. Errors concentrate in low-contrast arenas (Morris water maze, `mwm`), reflective scenes (`mirror`), and self-occluded postures (`elev`), rather than in a single architectural weakness.

Table 3. Single-task ablation and a cross-task variant on the same network (mean ± standard deviation over three seeds). Joint training preserves segmentation but loses a little pose accuracy; bidirectional cross-task guidance recovers part of the pose gap without closing it.

Configuration	mIoU	PCK@0.05
Seg-only	0.8089 ± 0.0009	—
Pose-only	—	0.9719 ± 0.0018
Joint MTL	0.8083 ± 0.0012	0.9605 ± 0.0027
+ Cross-task guidance	0.8081 ± 0.0018	$\mathbf{0.9667 \pm 0.0034}$

One minor confound is worth noting: in the pose-only configuration the segmentation head is switched off, so instance matching during training uses heatmap distance instead of mask IoU.

The Joint MTL configuration reaches an mIoU of 0.8083 ± 0.0012, essentially identical to the 0.8089 ± 0.0009 of the segmentation-only model. The difference (-0.0006) is well within the seed-to-seed variation, indicating that co-training neither improves nor harms segmentation accuracy.

For keypoint estimation, the Joint MTL variant reaches a PCK of 0.9605 ± 0.0027, a drop of 0.0114 from the 0.9719 ± 0.0018 of the pose-only baseline. Because this gap is larger than the seed-to-seed variation, it is a small but genuine case of negative transfer, or task interference [7,17]. Since the shared ResNet-34 encoder must serve two geometrically different tasks, it cannot fully reach the optimum of each one.

Can cross-task cues recover the gap? We also tested whether letting the heads exchange information, not just share a backbone, reduces this interference. We added a lightweight stop-gradient bidirectional link: the predicted mask gates each instance's keypoint channels, and the predicted heatmaps prime the segmentation decoder. Pose PCK then rises from 0.9605 to 0.9667 with segmentation unchanged (Table 3), recovering about half the gap to the pose-only ceiling. This mitigates the negative transfer but does not remove it, and the gain

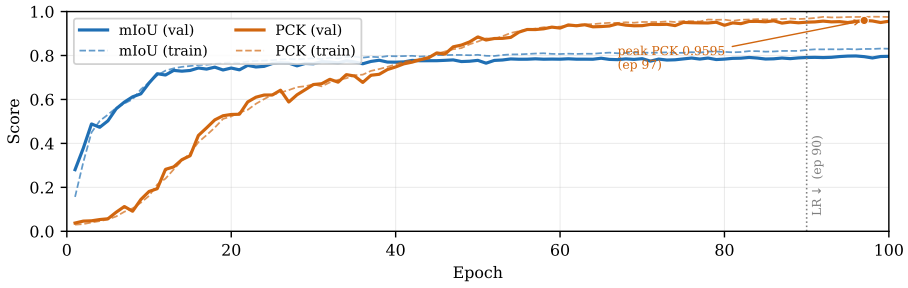


Fig. 4. Training dynamics of HybridMTLNet over 100 epochs (single 1:500 run). Segmentation mIoU converges quickly and saturates while pose PCK rises gradually; the dotted line marks the learning-rate decay at epoch 90, and the marker indicates the peak validation PCK (0.9595 at epoch 97).

is modest relative to the added design complexity. We therefore keep the plain joint model—the minimal unified architecture, and the clean setting in which we measure transfer—as our main result, treating the cross-task link as an optional enhancement rather than the headline. Fully closing the gap would need heavier gradient-coupled fusion of joint human parsing and pose [14,21], which conflicts with our real-time design and is left to future work.

The main advantage of HybridMTLNet is therefore not multi-task accuracy synergy but architectural unification and efficiency: a single pipeline that preserves segmentation accuracy at a small, partly recoverable cost in pose.

5.5 Training Dynamics

Figure 4 shows the training and validation curves over 100 epochs. The instance segmentation task converges quickly (training mIoU ≥ 0.78 by epoch 26) and then saturates, while keypoint PCK climbs more gradually. The ReduceLROnPlateau scheduler decays the learning rate at epoch 90 when the validation loss stops improving; the smaller steps then push validation PCK to its peak of 0.9595 at epoch 97 (a validation peak; the result tables report test-set scores). Small train–validation gaps (0.035 mIoU, 0.019 PCK) indicate good generalization.

5.6 Inference Efficiency

HybridMTLNet's efficiency comes directly from sharing one encoder. We compare it against a modular two-model baseline built from two single-task copies of the same architecture, which isolates the benefit of the shared encoder (Table 4). Running both tasks in a single pass, HybridMTLNet uses only 28.7 M parameters and 28.8 GFLOPs—about half the modular pipeline's 57.4 M / 57.5 GFLOPs—and reaches 111.8 FPS (8.9 ms/frame) on an NVIDIA RTX 5060 GPU (14.8 FPS on CPU) versus 55.9 FPS. As laboratory recordings typically run at 30–60 FPS, this comfortably supports real-time deployment in resource-constrained settings.

Table 4. Inference efficiency at 256×256 (batch size 1, averaged over 100 runs). The modular baseline runs two separate single-task models of the same architecture; GPU throughput is measured on an NVIDIA RTX 5060.

Approach	Params (M)	FLOPs (G)	GPU (FPS)	CPU (FPS)
Modular (2 models)	57.4	57.5	55.9	7.4
HybridMTLNet (Ours)	**28.7**	**28.8**	**111.8**	**14.8**

6 Conclusion

We propose HybridMTLNet, a unified box-free multi-task learning framework for simultaneous instance segmentation and keypoint estimation of laboratory mice in one forward pass. This lets us avoid bounding-box annotations, and we resolve instance-identity assignments with dynamic Hungarian matching on mask IoU. Our approach is based on a shared ResNet-34 encoder, with a dense U-Net-style decoder and a deconvolutional pose head.

Quantitative analysis shows that our unified model matches or exceeds the evaluated single-task baselines under our experimental setting, including YOLO26 and the single-run Mask R-CNN baseline. Our controlled ablation further shows that this segmentation quality derives mainly from the dense per-pixel architecture rather than multi-task synergy: single-task segmentation accuracy is preserved during joint training, at only a small, well-documented cost in pose accuracy. The real value of this work is thus its architectural unification and efficiency—halving parameters and computational complexity while doubling throughput on off-the-shelf hardware.

Our approach offers clear advantages, but two limitations point to future work. First, the fixed two-instance output head should be extended to variable, higher-density counts. Second, HybridMTLNet processes only static frames; adding temporal tracking to exploit cross-frame continuity would improve robustness for continuous video-based behavioral phenotyping.

Ethics Statement. The image data were originally collected by the Faculty of Pharmaceutical Sciences, Khon Kaen University, under approved animal-use protocols. This study performs only secondary computational analysis of these pre-existing images and involved no live-animal procedures.

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